

Analytical techniques

A Level Chemistry

Infrared spectroscopy

Infrared spectroscopy 红外光谱 helps you find the **functional group** 官能团 in a molecule. Each kind of bond soaks up (absorbs) infrared radiation at its own range of frequencies. Where the bond shows strong **absorption** 吸收, the spectrum has a dip.

The position is measured in **wavenumber** 波数 (in cm^{-1}). You are given a data table, so you do not memorise the numbers. You just match the dips to bonds:

- a broad dip around $3200\text{--}3650\text{ cm}^{-1}$ $\rightarrow$ an O–H bond in an alcohol.
- a dip around 1700 cm^{-1} $\rightarrow$ a C=O bond (aldehyde, ketone, acid or ester).
- a broad dip $2500\text{--}3000\text{ cm}^{-1}$ together with a C=O dip $\rightarrow$ a carboxylic acid.

This is useful for checking a reaction. For example, if propene has been turned into propan-2-ol, the C=C dip should be gone and an O–H dip should appear.

Mass spectrometry

In **mass spectrometry** 质谱, a molecule is turned into ions and sorted by its **mass-to-charge ratio** 质荷比 (m/e). The spectrum is a set of peaks at different m/e values.

Relative atomic mass from isotopes

An element's **isotope** 同位素 mixture gives several peaks. From the **isotopic abundance** 同位素丰度 (how common each isotope is) you can find the **relative atomic mass** 相对原子质量—a weighted average:

$$A_r = \frac{\sum(\text{isotope mass} \times \text{abundance})}{\sum \text{abundance}}$$

For example, chlorine is 75% ^{35}Cl and 25% ^{37}Cl , giving $A_r = \frac{35 \times 75 + 37 \times 25}{100} = 35.5$.

The molecular ion and fragmentation

The peak at the highest m/e (the **molecular ion** 分子离子 peak, or **molecular ion peak** 分子离子峰, M^+) gives the relative molecular mass of the whole molecule.

The molecule also breaks into smaller pieces —this is **fragmentation** 碎裂. The gap between two peaks tells you the mass of the lost piece, so you can suggest each **fragment** 碎片. For example, a loss of 15 means a CH_3 group was lost, and a loss of 29 means CHO or C_2H_5 .

The $[M + 1]$ and $[M + 2]$ peaks

- a small $[M + 1]$ peak comes from the ^{13}C isotope. The number of carbon atoms n is:

$$n = \frac{100 \times \text{abundance of } [M + 1]^+}{1.1 \times \text{abundance of } M^+}$$

- an $[M + 2]$ peak shows chlorine or bromine. One chlorine gives an $[M + 2]$ peak about one third the height of M^+ (from ^{37}Cl); one bromine gives an $[M + 2]$ peak about the **same** height as M^+ (from ^{81}Br).